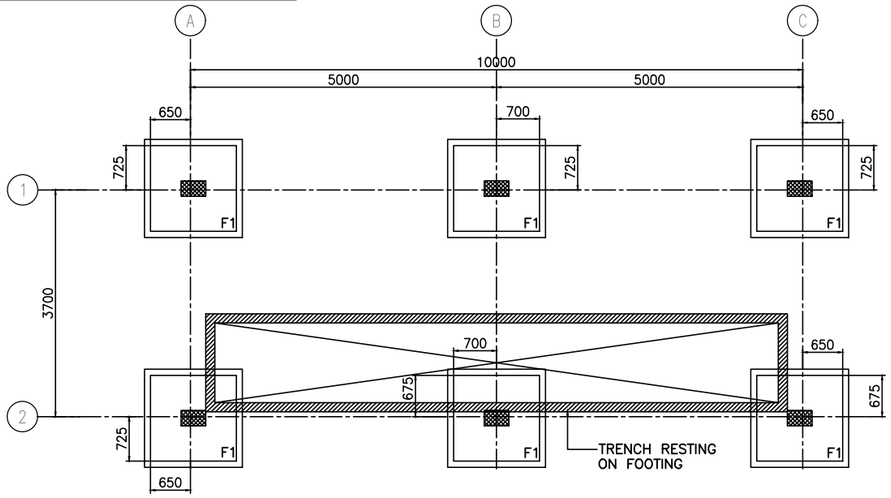
Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**GENERAL ARRANGEMENT DRAWING SECURITY BASE CAMP: ELECTRICAL ROOM LAYOUT & DETAILS OF BUILDING**
Document Number:	**MD-216-1800-EG-CV-GAD-1550**
Revision:	**D01**
Revision Date:	**20.02.2023**
Status:	**ASBUILT**

Outotec Document Number:	900707	GAU01	AN02	MTL01	18039	OU100676797	3
L&T Document Number:	O19045	900	C	G	24251	001	3

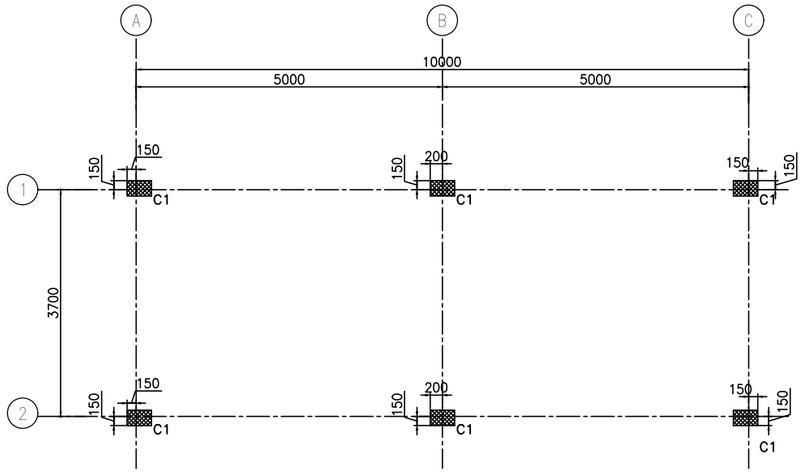
REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	13.10..2020	ISSUED FOR APPROVAL	----	AVG	SJM	
C01	16.11..2020	ISSUED FOR CONSTRUCTION	----	AVG	SJM	
C02	18.06..2020	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	----	AVG	SJM	
D01	20.02.2023	AS BUILT	----	AVG	SJM	

DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1550 | REV: D01



FOUNDATION LAYOUT
SCALE 1:50



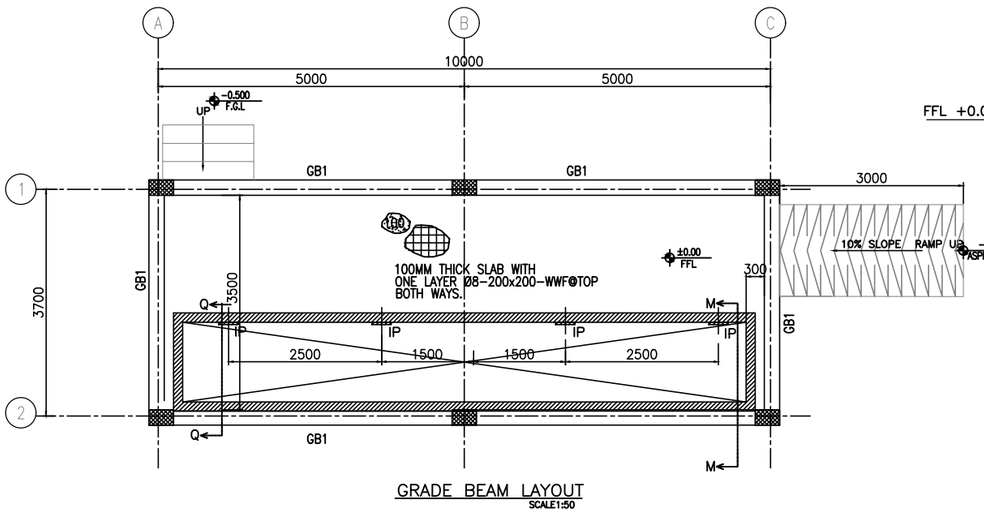
CONCRETE COLUMN LAYOUT
SCALE 1:50

SCHEDULE OF FOOTINGS

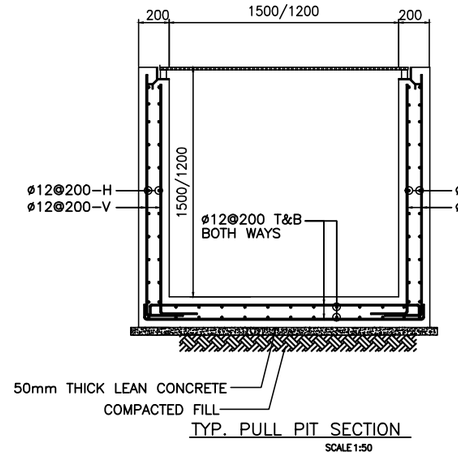
FOOTING MARK	PLAIN CONCRETE			REINFORCED CONCRETE			BOTTOM REINFORCEMENT		TOP REINFORCEMENT		Ha	REMARKS
	L	W	T	L	W	T	SHORT BARS	LONG BARS	SHORT BARS	LONG BARS		
F1	1600	1600	50	1400	1400	350	ø12@150	ø12@150	–	–	1500	

SCHEDULE OF GRADE BEAMS

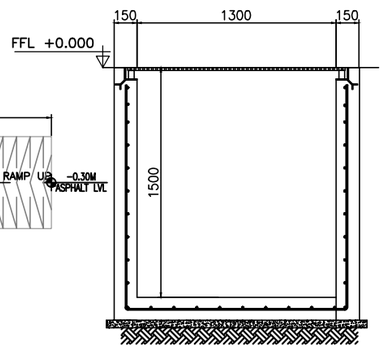
BEAM MARK	DIMENSIONS		STEEL REINFORCEMENT				STIRRUPS
			BOTTOM		TOP		
	WIDTH	DEPTH	STRAIGHT	ADD.	STRAIGHT	ADD.	
GB1	250	600	3ø16	–	3ø16	–	PROVIDE AT THE ØDISTANCE OF L/3 FROM THE SUPPORT IN ø10@140 IN THE END SPAN & MID SPAN



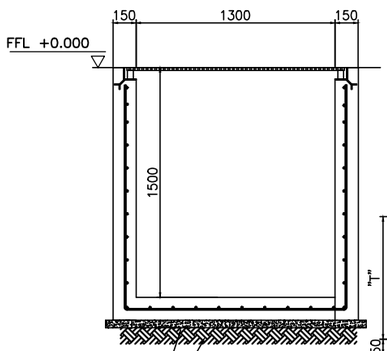
GRADE BEAM LAYOUT
SCALE 1:50



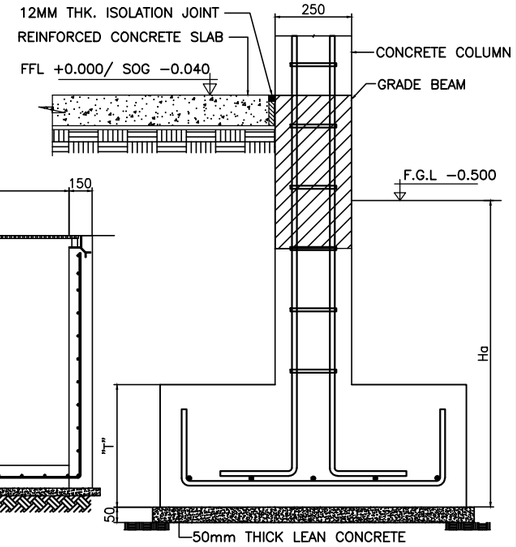
TYP. PULL PIT SECTION
SCALE 1:50



SECTION-Q
SCALE 1:20

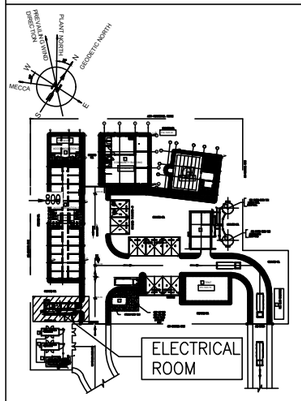


SECTION-M
SCALE 1:20

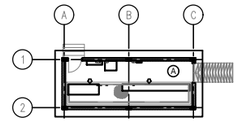


FOUNDATION SECTION
SCALE 1:20

MASTER KEY PLAN



ELECTRICAL ROOM ENLARGED KEYPLAN



ELECTRICAL ROOM			
BLDG NO.	CORNER NO.	EASTING	NORTHING
A	A1	E 262092156.02	N 2508695572.395
	C2	E 262102597.203	N 2508697733.81

NOTES:

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2. MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3. MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4. ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA .
5. THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6. WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7. THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a. CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b. CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND BEAMS, COLUMNS.....40MM SLABS, WALLS, JOISTS.....20MM
   c. CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9. F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 923.90 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

FOR DETAIL OF PULL PITS AND CONDUITS REFER DRAWING NOS. MD-216-1800-EG-CV-GAD-1578 & MD-216-1800-EG-EL-LAD-1005.

LEGENDS

F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL-LEVEL
RL-REDUCED LEVEL
V-VERTICAL
DOC-DOCUMENTS
SPECS-SPECIFICATIONS
BLDG.-BUILDING
Ø-DIA
THK.-THICK
IP-INSERT PLATE

SHT-1 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	20/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-AR-GAD-3172	BUILDING PLANS	C01	16/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	13/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: GULF CONSULT ARCHITECTS AND ENGINEERS

SC-ER-ST-001				REVISION A01	
DATE	03/10/2020	DATE	04/10/2020	DATE	05/10/2020
DRAWN	MAH	CHECKED	MF	APPROVED	FK

LARSEN & TOUBRO DRAWING NUMBER: 019045_900_C_G_24251_001 — REVISION 3

OUTOTEC DRAWING NUMBER: 900707__GAU01__AN02__MTL01__18039 — DOCUMENT ID: OU100676797 — REVISION 3

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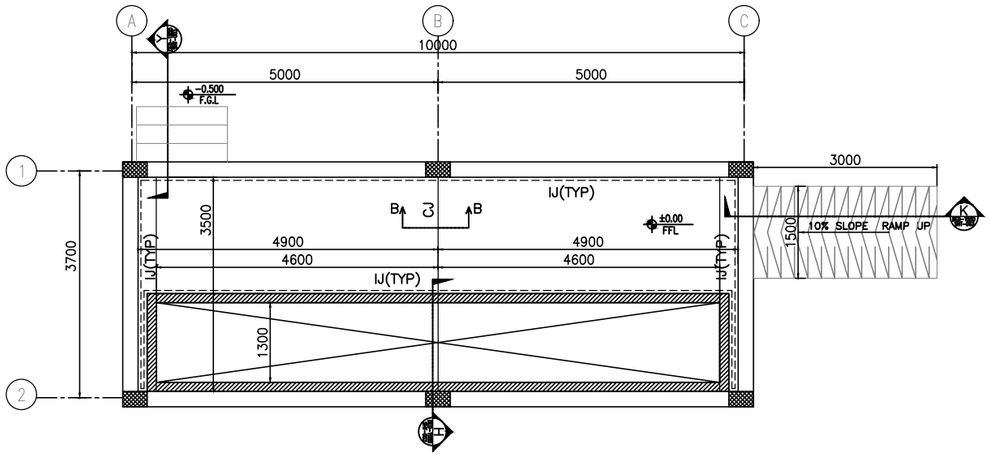
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Saudi Arabian Mining Company
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CHECKED	AVG	10/10/2020
ENGINEER	--	--
APPROVED	SJM	13/10/2020
EMPLOYER		
SCALE	1:50	A1

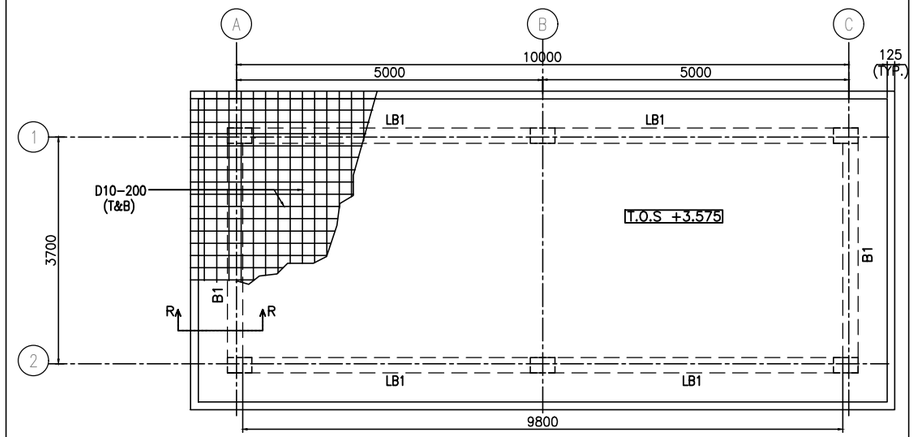
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECURITY BASE CAMP: ELECTRICAL ROOM
LAYOUT & DETAILS OF BUILDING

DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1550 — REV: D01

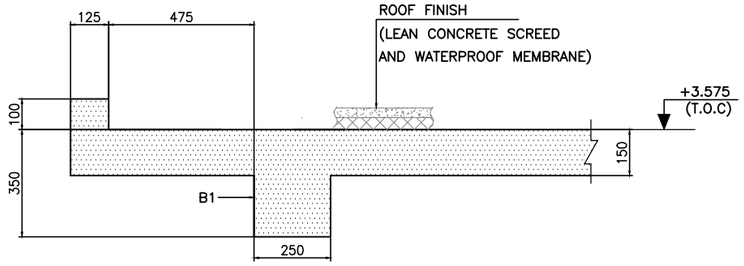
DRAWING NUMBER MD-216-1800-EG-CV-GAD-1550 REV D01



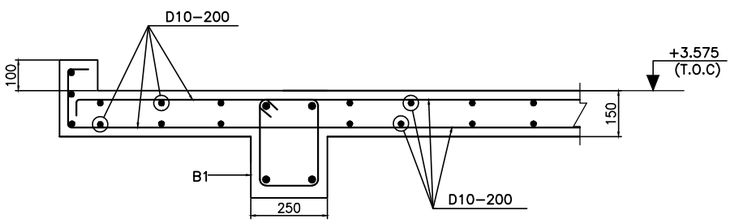
CONCRETE JOINTS LAYOUT
SCALE 1:50



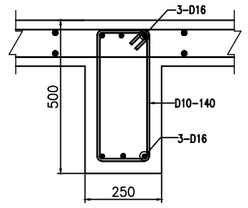
ROOF SLAB LAYOUT
SCALE 1:50



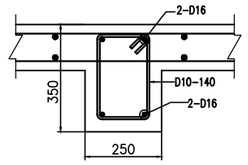
SECTION R-R
(SCALE 1:20)



(SHOWING RC DETAIL)
(SCALE 1:20)



R.C DETAIL OF BEAM MARKED LB1
(SCALE 1:10)

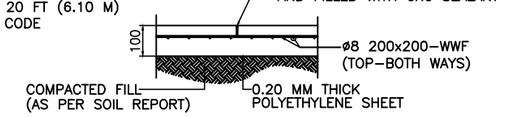


R.C DETAIL OF BEAM MARKED B1
(SCALE 1:10)

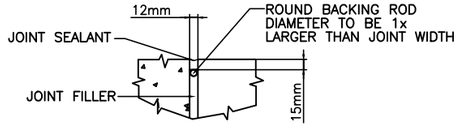
SCHEDULE OF ROOF BEAMS

BEAM MARK	DIMENSIONS		STEEL REINFORCEMENT				STIRRUPS	REMARKS
			TOP		BOTTOM			
	WIDTH	DEPTH	STRAIGHT	ADD.	STRAIGHT	ADD.		
LB1	250	500	3Ø16	–	3Ø16	–	PROVIDE AT THE @DISTANCE OF L/3 FROM THE SUPPORT IN ø10@140 IN THE END SPAN & MID SPAN	REFER DETAIL-X
B1	250	350	2Ø16	–	2Ø16	–	PROVIDE AT THE @DISTANCE OF L/3 FROM THE SUPPORT IN ø10@140 IN THE END SPAN & MID SPAN	REFER DETAIL-R

NOTE: CONTROL JOINTS SHALL BE PROVIDED AT MAXIMUM SPACING OF 20 FT (6.10 M) AS PER ACI CODE



SECTION-B (CJ)
SCALE NTS



TYP JOINT SEAL DETAIL C GAD-1550
SCALE NTS

NOTES:

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2. MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3. MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4. ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA .
5. THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6. WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7. THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a. CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b. CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
      BEAMS, COLUMNS.....40MM
      SLABS, WALLS, JOISTS.....20MM
   c. CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9. F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 923.90 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

FOR DETAIL OF PULL PITS AND CONDUITS REFER DRAWING NOS. MD-216-1800-EG-CV-GAD-1578 & MD-216-1800-EG-EL-LAD-1005.

LEGENDS

F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
I.I-ISOLATION JOINT
CJ-CONTROL JOINT
SCJ-SLAB CONSTRUCTION JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.(H)-HORIZONTAL
REINF.-REINFORCEMENT
LVL.-LEVEL
TOS-TOP OF SCREED
RL-REDUCED LEVEL
V-VERTICAL
DOC-DOCUMENTS
SPECS-SPECIFICATIONS
Ø-DIA
THK.-THICK
IP-INSERT PLATE

SHT-2 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	20/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	--	AVG	--	SJM	
MD-216-1800-EG-AR-GAD-3172	BUILDING PLANS	C01	16/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	13/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: مكتب الخليج للاستشارات الهندسية GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWING NUMBER						REVISION
SC-ER-ST-002						A01
DATE	03/10/2020	DATE	04/10/2020	DATE	05/10/2020	
DRAWN	MAH	CHECKED	MF	APPROVED	FK	

LARSEN & TOUBRO DRAWING NUMBER	REVISION
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OUTOTEC DRAWING NUMBER: 900707__GAU01__AN02__MTL01__18039	DOCUMENT ID: OU100676797 — REVISION: 3

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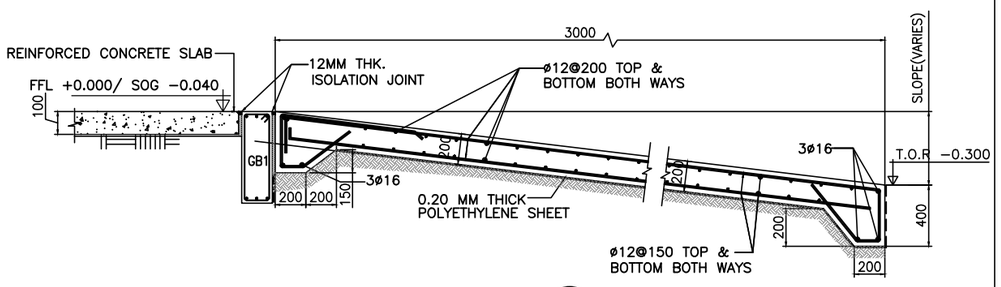
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CHECKED	AVG	10/10/2020
ENGINEER	--	--
APPROVED	SJM	13/10/2020
EMPLOYER		
SCALE	1:50	A1

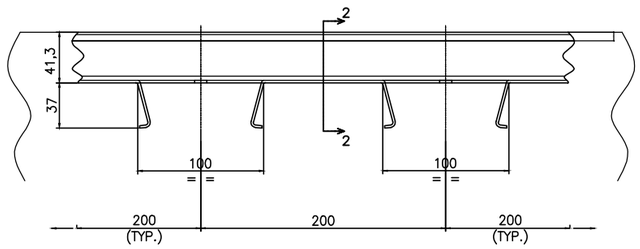
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECURITY BASE CAMP: ELECTRICAL ROOM
LAYOUT & DETAILS OF BUILDING

DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1550 REV: D01

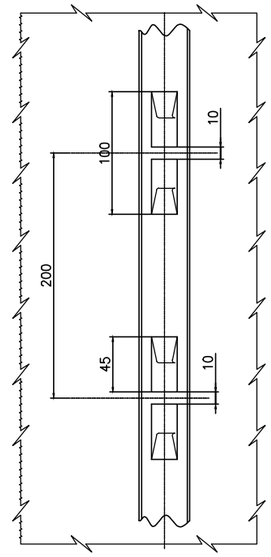
DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1550 | REV: D01



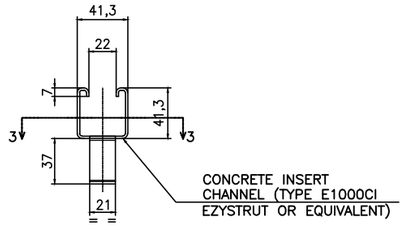
TYPICAL RAMP SECTION K
SCALE 1:20 GAD-1550 SHT-02



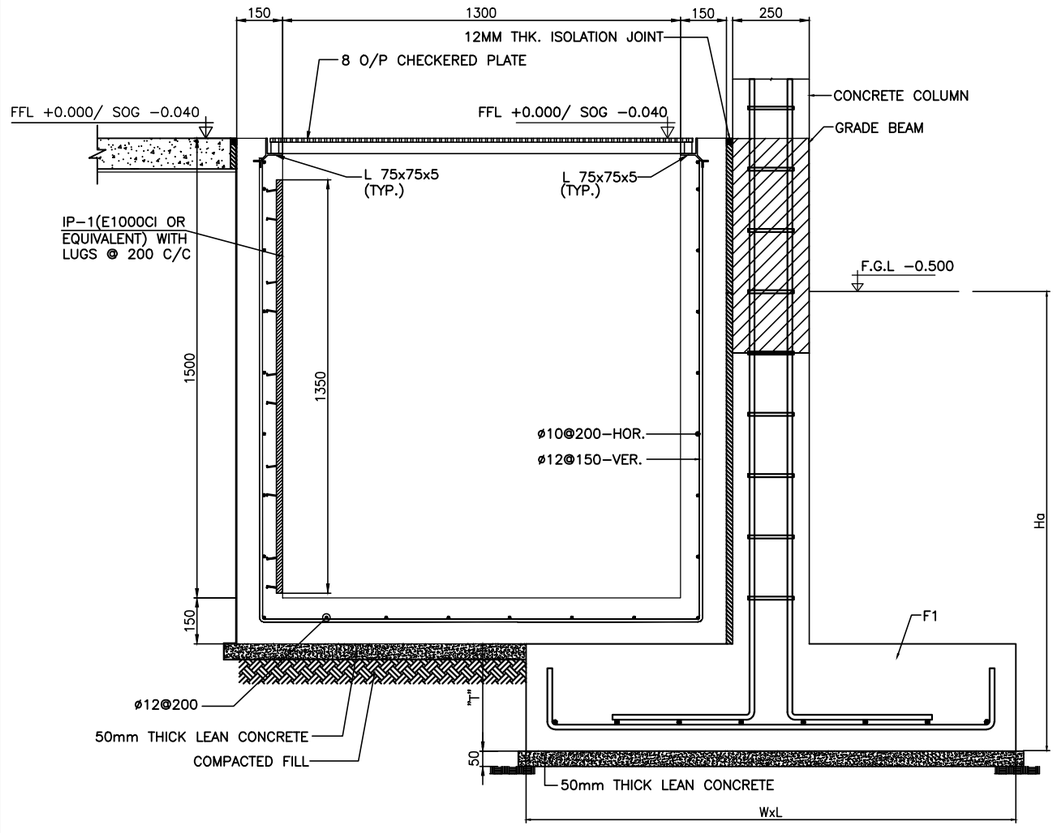
TYP. DET. OF INSERT MKD. IP−1 CHANNEL TYPE E1000CI OR EQUIVALENT
SCALE N.T.S



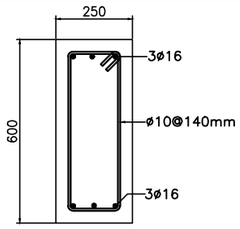
SECTION 3−3
SCALE N.T.S



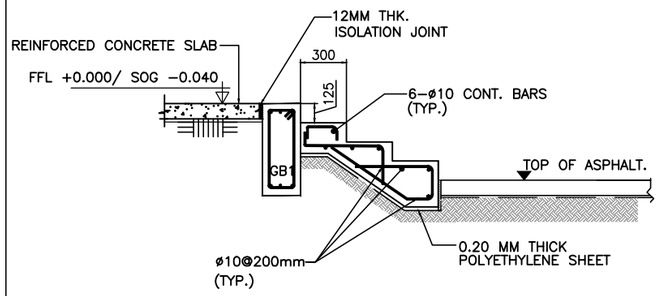
SECTION 2−2
SCALE N.T.S



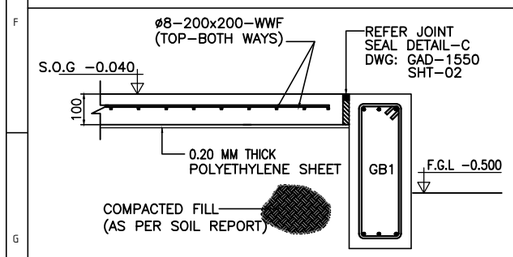
TYP. TRENCH SECTION H
SCALE 1:10 GAD-1550 SHT-02



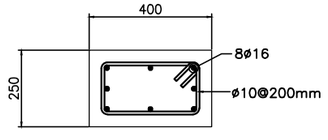
GB1 @ MIDSPAN & ENDSPAN S
SCALE 1:10 GAD-1550 SHT-02



TYP. STEP SECTION Y
SCALE 1:20 GAD-1550 SHT-02



TYP SECTION FOR GRADE SLAB −
SCALE 1:10 GAD-1550 SHT-02



COLUMN DETAIL C1
SCALE 1:10 GAD-1550 SHT-01

NOTES:

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2. MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS M35(5000 PSI).
3. MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS M15(2175 PSI).
4. ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA.
5. THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 350 KPA(REFER DOC 216-3000-EG-CV-CAL-1008).
6. WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECS AND ITS MATERIAL TO BE APPROVED BY ENGINEER ON SITE.
7. THE MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a. CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b. CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
      BEAMS, COLUMNS.....40MM
      SLABS, WALLS, JOISTS.....20MM
   c. CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9. F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 923.90 AS PER THE CIVIL DRAWING CE-004A-OVERALL LAYOUT PLAN.

FOR DETAIL OF PULL PITS AND CONDUITS REFER DRAWING NOS. MD-216-1800-EG-CV-GAD-1578 & MD-216-1800-EG-EL-LAD-1005.

LEGENDS

F-FOUNDATION
C-COLUMN
FFL-FINISH FLOOR LVL
FGL-FINISH GRADE LVL
TOS-TOP OF SCREED
WWF-WELDED WIRE FABRIC
HCS-HOLLOW CORE SLAB
GB-GRADE BEAM
IJ-ISOLATION JOINT
CJ-CONTROL JOINT
GA-GALVANIZED
S.O.G-SLAB ON GRADE
ARCH-ARCHITECTURAL
HOR.-HORIZONTAL
REINF.-REINFORCEMENT
LVL.-LEVEL
RL-REDUCED LEVEL
VER-VERTICAL
O/P-OVER PLATE
T.O.R-TOP OF ROAD
Ø-DIA
THK.-THICK
IP-INSERT PLATE

SHT-3 OF 3

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	20/02/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	22/06/2021	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-AR-GAD-3172	BUILDING PLANS	C01	16/11/2020	ISSUED FOR CONSTRUCTION	--	AVG	--	SJM	
MD-216-6000-EG-CV-GAD-1535	OVERALL LAYOUT PLAN	A01	13/10/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: مكتب الخليج للاستشارات الهندسية — GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWING NUMBER					REVISION
SC-ER-ST-003					A01
DATE	03/10/2020	DATE	04/10/2020	DATE	05/10/2020
DRAWN	MAH	CHECKED	MF	APPROVED	FK

LARSEN & TOUBRO DRAWING NUMBER	REVISION
019045_900_C_G_24251_001	3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AN02__MTL01__18039	DOCUMENT ID: OU100676797 — REVISION: 3

LARSEN & TOUBRO

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DRAWN	--	--
CHECKED	AVG	10/10/2020
ENGINEER	--	--
APPROVED	SJM	13/10/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECURITY BASE CAMP: ELECTRICAL ROOM
LAYOUT & DETAILS OF BUILDING

DRAWING NUMBER: MD-216-1800-EG-CV-GAD-1550 | REV: D01